# Léon MORALES

**Generative AI Engineer — LLM & Graph RAG**

Available from April 2027 • Open to relocation worldwide

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## PROFILE

Dual-degree engineer (UTBM / UQAC) specializing in Generative AI and Machine Learning, with a strong interest in Large Language Models (LLMs) and Deep Learning. Experience in R&D (drug discovery, academic research) and in designing RAG (Retrieval-Augmented Generation) systems on knowledge graphs.

## EXPERIENCE

**AI & Data Innovation Engineer — End-of-studies internship** Sept. 2026 – Feb. 2027
*Astek Group • confidential project* La Défense, France

- Designing a **Graph RAG** pipeline in **Python**: leveraging a **Neo4j** knowledge graph to provide relevant context to an LLM.
- Deploying local LLMs with **Ollama** and generating answers enriched with context retrieved from the graph.
- RAG evaluation (RAGAS, LLM-as-a-judge).
- Agentic approach to information retrieval in the graph.

**Academic research project — Human Activity Recognition** Jan. – May 2026
*LIARA Lab (Ambient Intelligence), UQAC* Chicoutimi, Canada

- Designed a self-supervised **contrastive learning** architecture for wearable sensors (IMU/PPG), aiming to outperform state-of-the-art involution networks.
- Full research cycle: literature review, implementation (PyTorch), experimental validation; scientific paper submitted for publication.

**AI Engineering Intern** Sept. 2024 – Feb. 2025
*KSILINK* Strasbourg, France

- **Drug Discovery R&D:** implemented a multimodal contrastive learning model (cell images and molecular structures), evaluated with several similarity metrics, based on *Cell Morphology-Guided Small Molecule Generation with GFlowNets*.
- **Graph RAG chatbot:** designed and deployed a conversational agent linking an open-source LLM to a Neo4j database (100k+ nodes), enabling about fifteen biologists to query their knowledge base in natural language.
- **Computer Vision:** trained and evaluated CNNs (PyTorch Lightning) for cell image classification on noisy datasets.

**President of the SKI'UTBM Association** July 2023 – Feb. 2024
*UTBM* Belfort, France

- Ski trip for 200 students: managed a €100k+ budget and a team of 11 people.

## PROJECTS

**RAG Assistant & Dashboard** — *Conformit X UQAC Hackathon, 2nd place (team of 4, $1,000)*, Nov. 2025
AI assistant for industrial risk management built in 25 hours: RAG pipeline translating natural language into secure SQL. *Python, Gemini, SQL, FastAPI, Streamlit, Docker.* Source code.

**Graph RAG Chatbot (Neo4j) & AI Portfolio** — website, source code
Graph RAG chatbot on a Neo4j database, embedded in my portfolio alongside a CNN model (ONNX).

## EDUCATION

**Master's Degree in Computer Science (AI profile) — Dual degree** 2025 – 2027 (expected)
*Université du Québec à Chicoutimi (UQAC)* Chicoutimi, Canada

**Master of Engineering in Computer Science — Data Science & AI** 2021 – 2027 (expected)
*Université de Technologie de Belfort-Montbéliard (UTBM)* Belfort, France

**International Exchange Semester — Computer Engineering (in English)** Jan. – June 2023
*Yeditepe University* Istanbul, Türkiye

## SKILLS

- **Generative AI & LLMs:** LLMs (Ollama, Hugging Face Transformers), RAG / Graph RAG, LangChain, embeddings, vector databases (ChromaDB, Neo4j vector index), fine-tuning
- **Machine Learning & Data:** PyTorch, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Deep Learning, NLP, Computer Vision (CNN)
- **Languages & tools:** Python (advanced), SQL (PostgreSQL), Cypher, C/C++, Java; Git/GitHub, Docker, Neo4j, FastAPI, Streamlit, Weights & Biases
- **Spoken languages:** French (native), English (C1, professional), Spanish (B1)

## INTERESTS

Travel (15+ countries) • Guitar (7 years) • Reading • Sports: skiing, basketball, snowboarding, ice skating, swimming